

Notes: Foucault, Hombert, and Roşu (JF, 2016)

News Trading and Speed

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1 Big picture

- **Question.** How does speed change the trading strategy of an informed speculator when his signals forecast both the long-run value of an asset and the short-run price reaction to incoming news?
- **Setting.** The paper builds a dynamic Kyle-style model with one informed speculator, one competitive dealer, and noise traders. The dealer receives public news; the speculator receives private signals correlated with that news.
- **Core puzzle.** Empirically, directional high-frequency traders' aggressive orders are strongly related to news and very short-run returns, yet a large share of their profits seems to come from longer-horizon price changes. A pure “trade only on immediate news” story is therefore incomplete.
- **Main theoretical result.** *News trading* arises only when the speculator is **fast**, meaning he can trade before the dealer updates quotes to reflect incoming news. In that case, the speculator's optimal trade has two components:
 - a **value-trading** component, driven by the gap between his estimate of long-run value and the dealer's quote;
 - a **news-trading** component, driven by his forecast of very short-run price reactions to imminent news.
- **Why speed matters.** When the speculator is fast, his position becomes much more volatile, his trades account for a much larger share of trading volume, and his orders are more tightly linked to short-horizon returns. But he still earns an important part of his profits from longer-horizon value trading.
- **Main market-quality result.** Speed makes the market **less liquid** in the sense that prices react more to trades. However, speed does *not* change the overall speed of price discovery in the model: trades become more informative about short-run innovations but less informative about long-run mispricing, and the two effects offset.
- **Main comparative-statics result.** If news is more informative, a fast speculator trades more aggressively on news, contributes more to trading volume, and gains relatively more from short-run bets. Surprisingly, average liquidity can *improve* at the same time because the dealer learns more from better news.
- **Why this matters.** The model gives a coherent explanation for several stylized facts about directional HFTs: high participation rates, strong short-run forecasting ability, many round-trip trades, but profits that are not exclusively ultra-short-term.
- **Economic takeaway.** Speed does not merely let traders process the same information faster. It changes *what it is optimal to trade on*. Once a trader can get ahead of price adjustment to news, short-run news trading becomes a distinct and important equilibrium activity.

2 Model setup and key objects

2.1 Asset value and information flow

- Trading takes place continuously on the interval $[0, 1]$.
- The risky asset's liquidation value is

$$v_1 = v_0 + \int_0^1 dv_t, \quad dv_t = \sigma_v dB_t^v,$$

where B_t^v is a Brownian motion, $\sigma_v > 0$, and

$$v_0 \sim N(0, \Sigma_0).$$

- The paper interprets v_1 as the asset's **long-run** value, for example the end-of-day value. This is “long run” relative to the very short-run price movements generated by news.
- At date 0, the speculator observes v_0 . So Σ_0 measures his initial informational advantage.
- Over each infinitesimal interval $[t, t + dt]$, the speculator privately observes the innovation dv_t .
- The dealer receives public news

$$dz_t = dv_t + de_t, \quad de_t = \sigma_e dB_t^e,$$

where B_t^e is an independent Brownian motion.

- **News informativeness.** News is more informative when σ_e is smaller. Equivalently, the paper uses

$$\nu = \frac{1}{\sigma_e}$$

as a measure of news informativeness.

- **Key correlation.** The speculator's signal is informative about both:
 - the long-run asset payoff, and
 - the short-run public news that will soon move prices.
- This is the primitive force behind the paper's mechanism: the same private signal can support both a long-run valuation bet and a very short-run news bet.

2.2 Traders, order flow, and pricing

- There are three market participants:
 - one risk-neutral informed speculator,
 - competitive risk-neutral dealer,
 - noise traders.
- Noise-trader order flow is

$$du_t = \sigma_u dB_t^u,$$

where B_t^u is independent of all other shocks.

- The total order flow executed by the dealer is

$$dy_t = dx_t + du_t.$$

- The dealer's information set at time t is

$$I_t = \{y_\tau\}_{\tau \leq t} \cup \{z_\tau\}_{\tau \leq t}.$$

- The dealer's quote is her conditional expectation of final value:

$$q_t = \mathbb{E}(v_1 \mid I_t).$$

- Thus the dealer learns from two sources:

- public news dz_t ,
- informative order flow dy_t .

2.3 Fast versus slow timing

The paper's central distinction is the timing of trade relative to news incorporation.

- **Fast speculator model.** The speculator trades *before* the dealer updates her quote to reflect the current news. Execution takes place at

$$p_{t+dt} = \mathbb{E}(v_1 \mid I_t \cup dy_t).$$

- **Slow speculator model.** The dealer first incorporates news and then executes the order flow:

$$p_{t+dt} = \mathbb{E}(v_1 \mid I_t \cup dz_t \cup dy_t).$$

- In words:
 - **fast** = can get ahead of the dealer's news reaction,
 - **slow** = cannot.
- This timing difference is the entire source of the new economics. All primitives are otherwise the same.

2.4 Trading strategy, profits, and equilibrium notion

- Let x_t be the speculator's position. His expected continuation profit from date τ onward is

$$\pi_\tau = \mathbb{E} \left(\int_\tau^1 (v_1 - p_{t+dt}) dx_t \mid J_\tau \right),$$

where J_t is the speculator's information set.

- The paper focuses on linear equilibria in which the speculator's trade takes the form

$$dx_t = \beta_t^k (v_t - q_t) dt + \gamma_t^k dv_t, \quad k \in \{S, F\}.$$

- **Value-trading component:**

$$\beta_t^k(v_t - q_t)dt.$$

This is the standard Kyle-type bet on long-run mispricing.

- **News-trading component:**

$$\gamma_t^k dv_t.$$

This is a bet on the short-run price reaction to imminent news.

- **Definition of news trading.** The paper says there is *news trading* when $\gamma_t^k > 0$, so that the speculator's signal affects his trade above and beyond its effect through long-run valuation.

Interpretation. The decomposition is the heart of the paper. The drift part slowly builds the desired long-run position. The stochastic part captures rapid, round-trip trading on soon-to-arrive news.

3 Models and theoretical results (all equations + interpretation)

3.1 Benchmark slow equilibrium

Theorem 1. In the slow model, the unique linear equilibrium is

$$dx_t = \beta_t^S(v_t - q_t)dt + \gamma^S dv_t,$$

$$p_{t+dt} = q_t + \mu^S dz_t + \lambda^S dy_t,$$

$$dq_t = \mu^S dz_t + \lambda^S dy_t,$$

with

$$\gamma^S = 0.$$

- The slow speculator has **no** news-trading component.
- His trade depends only on long-run mispricing, exactly in the spirit of Kyle (1985).
- Prices respond to both:
 - news directly through $\mu^S dz_t$,
 - order flow through $\lambda^S dy_t$.

Interpretation. When the dealer updates the quote before the speculator can trade, the short-run price reaction to news is already gone. So there is nothing to exploit with a separate news bet.

3.2 Fast equilibrium

Theorem 2. In the fast model, the unique linear equilibrium is

$$dx_t = \beta_t^F(v_t - q_t)dt + \gamma^F dv_t,$$

$$p_{t+dt} = q_t + \lambda^F dy_t,$$

$$dq_t = \lambda^F dy_t + \mu^F(dz_t - \rho^F dy_t),$$

where the coefficients are functions of the primitive parameters and of a scalar $g \in (0, 1)$ solving

$$g = \left(1 + \frac{\sigma_e^2}{\sigma_v^2}g\right) \frac{(1+g)^2}{\left(2 + \frac{\sigma_e^2}{\sigma_v^2} + \frac{\sigma_e^2}{\sigma_v^2}g\right)^2} \frac{\sigma_v^2}{\sigma_v^2 + \Sigma_0}.$$

- Now $\gamma^F > 0$, so the speculator trades on pending news.
- The transaction price reacts only to total order flow at execution, because the dealer has not yet incorporated current news at the instant of trade.
- The dealer's subsequent quote update uses both:
 - order flow,
 - the *surprise component* of news, $dz_t - \rho^F dy_t$.
- Intuitively, because the speculator trades before news is processed, order flow itself reveals something about the news that is about to hit the market.

Interpretation. The fast trader partially front-runs the dealer's news update. The dealer therefore learns about current news from the trade itself, and later reacts only to the part of public news not already inferred from order flow.

3.3 Core speed comparisons

3.3.1 Proposition 1: news trading requires speed

Proposition 1.

$$\gamma^F > 0, \quad \gamma^S = 0.$$

- News trading exists only when the speculator is fast.
- Speed is therefore not just a matter of earning more from the same strategy; it creates a *new equilibrium margin of trade*.

3.3.2 Proposition 2: fast traders lean less on long-run valuation

Proposition 2. For all dates and parameter values,

$$\beta_t^F < \beta_t^S.$$

- A fast trader exploits each signal more aggressively overall because he adds a news bet.
- That causes his long-run informational advantage to be incorporated into prices more quickly.
- To avoid dissipating his informational edge too fast, he scales down the pure value-trading component.

3.3.3 Proposition 3: profits

Let π_β^k denote expected profit from value trading and π_γ^k expected profit from news trading.

Proposition 3.

$$\begin{aligned} \pi_\beta^k > 0 \quad \text{for } k \in \{S, F\}, \quad \pi_\beta^S > \pi_\beta^F > 0, \\ \pi_\gamma^F > 0, \quad \pi_\gamma^S = 0, \quad \pi_0^F > \pi_0^S. \end{aligned}$$

- Both slow and fast traders make money from long-run information.
- Only fast traders make money from short-run news trading.
- Total expected profit is higher when the trader is fast, but *value-trading* profit is actually lower.

Important implication. A trader can look highly “short-term” in his order flow and still earn most of his money from longer-horizon bets.

3.3.4 Propositions 4 and 5: liquidity and quote sensitivity**Proposition 4.**

$$\lambda^F > \lambda^S.$$

- Price impact is higher when the speculator is fast.
- The dealer is more exposed to selling just before good news and buying just before bad news.

Proposition 5.

$$\mu^F < \mu^S.$$

- When the speculator is fast, order flow already contains information about the impending news.
- So the surprise component of public news is smaller, and the dealer reacts less to news itself.

Interpretation. Speed shifts informational content *from public news to trades*. Prices react more to trades and less to subsequent news.

3.3.5 Corollaries 1 and 2: volatility decomposition and price discovery

Corollary 1. Instantaneous price volatility is the same regardless of speed:

$$\frac{1}{dt} \text{Var}(dq_t) = \sigma_v^2 + \Sigma_0.$$

- Total volatility is unchanged.
- But trades account for a larger fraction of volatility when the speculator is fast.

Corollary 2. Pricing efficiency is unchanged across the two models:

$$\Sigma_t = \mathbb{E}[(v_t - q_t)^2] = (1 - t)\Sigma_0.$$

- Fast trading makes order flow more informative about *short-run* innovations.
- But it makes order flow less informative about *long-run* mispricing.
- In equilibrium, the two effects offset exactly.

Interpretation. Speed reallocates where information shows up in prices, but not the overall rate at which the dealer’s pricing error declines.

3.3.6 Corollary 3: profits remain even with tiny initial information advantage

Corollary 3. As $\Sigma_0 \rightarrow 0$, the speculator still earns strictly positive expected profit from value trading whenever $\sigma_e > 0$, and if he is fast he also earns strictly positive expected profit from news trading.

- Even a vanishing initial informational edge does not kill the mechanism, because the trader keeps receiving new signals over time.
- The result is especially strong in continuous time: the trader reacts extremely quickly to fresh information.

3.4 Footprints of fast news trading

3.4.1 Participation rate

The instantaneous speculator participation rate is

$$\text{SPR}_t = \frac{\text{Var}(dx_t)}{\text{Var}(dy_t)} = \frac{\text{Var}(dx_t)}{\text{Var}(du_t) + \text{Var}(dx_t)}.$$

Corollary 4.

$$\text{SPR}^S = 0, \quad \text{SPR}^F = \frac{g}{1+g} > 0.$$

- Over infinitesimal intervals, the slow speculator contributes essentially nothing to volume.
- The fast speculator contributes a strictly positive share because his position changes are dominated by the stochastic news-trading component.

3.4.2 Autocorrelation of trades

Corollary 5. Over very short intervals, trades are less autocorrelated when the speculator is fast:

$$\text{Corr}(dx_t^S, dx_{t+\tau}^S) > 0, \quad \text{Corr}(dx_t^F, dx_{t+\tau}^F) = 0.$$

- Value trading naturally creates positively autocorrelated orders.
- News trading is driven by fresh innovations, so it does not persist in the same way.

3.4.3 Cumulative price impact at horizon τ

The paper defines the cumulative price impact of the speculator's trade over horizon τ as

$$\text{CPI}_t^k(\tau) = \frac{\text{Cov}(dx_t, q_{t+\tau} - q_t)}{dt}, \quad k \in \{S, F\}.$$

Corollary 6. As $\tau \rightarrow 0$:

- $\text{CPI}_t^S(\tau) \rightarrow 0$ in the slow model,
- $\text{CPI}_t^F(\tau)$ stays strictly positive in the fast model.

Interpretation. A positive very-short-horizon correlation between trades and subsequent returns is a diagnostic footprint of news trading.

3.5 News informativeness

The paper studies cross-sectional variation in

$$\nu = \frac{1}{\sigma_e}.$$

3.5.1 Proposition 6

Proposition 6. When the speculator is fast,

$$\frac{\partial \gamma^F}{\partial \nu} > 0.$$

Regardless of whether he is fast or slow,

$$\frac{\partial \beta_t^k}{\partial \nu} < 0 \quad \text{for } k \in \{S, F\}.$$

- More informative news makes short-run price reactions larger.
- So fast traders scale up news trading.
- But they scale down long-run value trading because information gets incorporated into prices faster.

If $\nu = 0$ (uninformative news), then

$$\gamma^F = 0 \quad \text{and} \quad \beta_t^F = \beta_t^S.$$

Interpretation. Both ingredients are necessary for news trading:

- **speed**, and
- **predictable price reaction to news**.

3.5.2 Corollary 7: more informative news can raise both trading and liquidity

Corollary 7. When the speculator is fast, higher news informativeness increases:

- the speculator's participation rate,
- total trading volume per unit of time,
- average liquidity, measured by $1/\lambda^F$.

Interpretation.

- This is the paper's most counterintuitive comparative statics result.
- Better news makes the fast trader more active, which tends to worsen adverse selection.
- But it also makes the dealer better informed about long-run value, which tends to improve liquidity.
- In the model, the second effect dominates on average.

3.5.3 Corollaries 8 and 9: profits and the incentive to become fast

Corollary 8. For a fast speculator:

- news-trading profits rise with ν ,
- value-trading profits fall with ν ,
- total expected profit falls with ν .

Corollary 9. The *gain from being fast*,

$$\pi_0^F(\nu) - \pi_0^S(\nu),$$

increases with news informativeness.

Interpretation.

- More informative news reduces total informational rents because the dealer learns more.
- But it increases the *relative* value of speed, since only a fast trader can monetize the larger short-run reaction of prices to news.

3.6 Extension: infrequent news and dealer latency

The extension allows trading to be more frequent than news.

- Each interval $[t, t + dt]$ is split into m trading rounds.
- The dealer receives news only after $\ell \leq m$ rounds.
- Thus:
 - m = trading frequency relative to news frequency,
 - ℓ = dealer latency.

The dealer's transaction price in round (t, j) is

$$p_{t+dt,j} = \begin{cases} \mathbb{E}(v_1 \mid I_t \cup dy_{t,1} \cup \dots \cup dy_{t,j}), & 1 \leq j \leq \ell, \\ \mathbb{E}(v_1 \mid I_t \cup dy_{t,1} \cup \dots \cup dy_{t,j} \cup dz_t), & \ell + 1 \leq j \leq m. \end{cases}$$

The speculator's linear strategy is

$$dx_{t,j} = \beta_{t,j}(v_t - q_t)dt + \gamma_{t,j}(dv_t - dw_{t,j}), \quad j = 1, \dots, m,$$

where $dw_{t,j}$ is the dealer's expectation of the incoming news based on prior rounds.

- If $\ell = 0$, the trader is slow and there is again no news trading.
- If $\ell \geq 1$, the trader is fast and can trade ahead of the news release.
- In the fast case, illiquidity, news-trading intensity, and participation are higher *before* news arrival than after.

Interpretation. The extension preserves the baseline intuition and adds an important timing prediction: the market is especially vulnerable to adverse selection immediately before scheduled news is incorporated.

4 Mechanism and relation to the literature (what makes the results work)

4.1 Why speed is essential

- The speculator's signal forecasts incoming news.
- Incoming news moves prices.
- Therefore the signal forecasts short-run price changes.
- But the trader can monetize this forecast only if he can trade before the dealer reacts.

Bottom line. Without a timing edge, the short-run bet disappears.

4.2 Why a fast trader can look short-term but still profit long-term

- Over short horizons, changes in the fast trader's position are dominated by the news-trading term $\gamma^F dv_t$.
- So his orders look highly correlated with immediate returns and with news.
- Over longer horizons, his position still drifts with the valuation term $\beta_t^F(v_t - q_t)dt$.

Bottom line. High-frequency order flow does not imply exclusively high-frequency profits.

4.3 Why liquidity falls but price discovery does not

- Liquidity falls because the dealer faces more adverse selection on short-run news.
- Price discovery does not change because:
 - trades reveal more about fresh innovations,
 - trades reveal less about the remaining long-run pricing error.

Bottom line. Speed changes the *composition* of information in trades, not the total amount of price discovery.

4.4 Why more informative news can raise both informed trading and average liquidity

- In most public-information models, better public information crowds out informed trading.
- Here, better news also raises the payoff to getting ahead of the short-run price reaction to that news.
- This extra short-run motive is unique to the fast model.

Bottom line. The paper's unusual comparative statics comes from combining public-information effects with a speed advantage.

4.5 Relation to other Kyle-type models

- **Back and Pedersen (1998)**. Dealers receive no news. In the language of this paper, this is like $\nu = 0$, so there is no news trading.
- **Chau and Vayanos (2008), Li (2013)**. Dealers incorporate news before informed traders can act. That corresponds to the *slow* case.
- **Foster and Viswanathan (1990)**. Dealers get news, but the news arrival rate is effectively negligible relative to trading opportunities, so there is no meaningful trade-ahead-of-news margin.
- **Martinez and Roşu (2013)**. Informed-trader positions can have a volatility component, but it comes from ambiguity aversion rather than from forecasting short-run price reactions to public news.
- **Huddart, Hughes, and Levine (2001) and Cao, Ma, and Ye (2013)**. Disclosure requirements can also generate noisy informed trades, but the stochastic component there is white noise and does not forecast news or short-run returns.

Interpretation. The paper’s distinctive contribution is not just a continuous-time Kyle extension. It is the combination of **gradual information**, **public news**, and a **timing advantage**.

5 Tables and figures

5.1 Figure 1: fast and slow speculation

Read:

- The left panel shows that the fast trader’s inventory path is much more volatile than the slow trader’s.
- The right panel shows many rapid buys and sells for the fast trader, consistent with frequent round trips.
- The point is not simply that the fast trader trades more. He trades in a *different way*: short-run inventory changes are dominated by reactions to incoming news.
- This is the visual summary of the paper’s core mechanism.

5.2 Figure 2: timing of events in the baseline model

Read:

- The entire difference between the fast and slow models is the order of two events:
 - does the trader submit the order before the dealer incorporates news?
 - or after?
- The figure makes clear why the paper can attribute changes in equilibrium behavior specifically to speed.
- It also makes clear why news trading is impossible in the slow model.

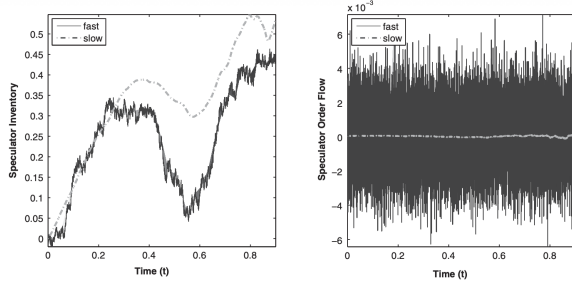


Figure 1. Fast and slow speculation. The figure plots the evolution over time of the speculator's position (left graph) and the change in this position—the speculator's trade—(right graph) when he is fast (plain line) and when he is slow (dot-dashed line) using the characterization of his equilibrium trading strategy in each case for the following parameter values (see Section II): $\sigma_u = 1$ (standard deviation of noise traders' order flow), $\sigma_v = 1$ (standard deviation of innovations in the asset value), $\sigma_e = 1$ (standard deviation of noise in news) and $\Sigma_0 = 1$ (variance of asset value conditional on information available at date 0). The path for the signals received by the speculator and the dealer is the same in each case. News and trades are assumed to take place every second. The liquidation time $t = 1$ corresponds to one trading day = 23,400 seconds.

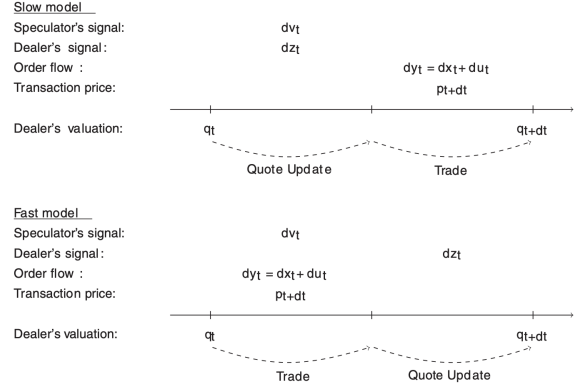


Figure 2. Timing of events during $[t, t + dt]$.

5.3 Table I: participation rates at different sampling frequencies

Read:

- At the **1 millisecond** horizon, the model implies:
 - $SPR^{slow} \approx 0.000008\%$,
 - $SPR^{fast} \approx 19.8238\%$.
- At the **1 minute** horizon:
 - $SPR^{slow} \approx 0.4615\%$,
 - $SPR^{fast} \approx 20.0776\%$.
- So the fast trader dominates trading volume at very short horizons, while the gap narrows only when the econometrician aggregates trades over longer windows.
- This is exactly the kind of sampling-frequency prediction that connects the model to HFT data.

Table I
Speculator's Participation Rate at Various Sampling Frequencies

The table compares the speculator participation rate (SPR) in the slow (SPR^{slow}) and fast (SPR^{fast}) models when data are sampled at various frequencies, as explained in Internet Appendix II. The order flow is thus aggregated—for the speculator and the noise traders—over time intervals of various lengths: one millisecond (10^{-3} seconds), 10^{-1} seconds, one second, and one minute. The parameter values are $\sigma_u = 1$ (standard deviation of noise traders' order flow), $\sigma_v = 1$ (standard deviation of innovations in the asset value), $\sigma_e = 0.5$ (standard deviation of noise in news), and $\Sigma_0 = 0.25$ (variance of asset value conditional on information available at date 0). The liquidation time $t = 1$ corresponds to one trading day = 23,400 seconds.

Sampling Interval	10^{-3} second	10^{-1} second	One second	One minute
SPR^{slow}	0.000008%	0.0008%	0.0077%	0.4615%
SPR^{fast}	19.8238%	19.8242%	19.8280%	20.0776%
SPR^{fast}/SPR^{slow}	2,577.096	25,771.51	2,577.65	43.50

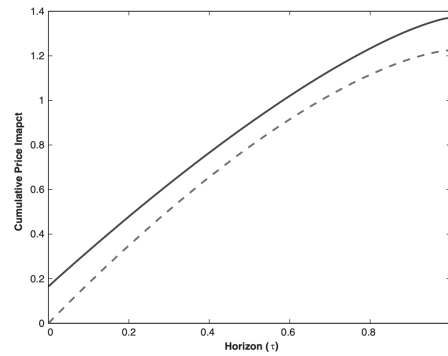


Figure 3. Cumulative price impact at different horizons. The figure plots the cumulative price impact, $CPI_0(\tau)$, over an interval of length $\tau \in (0, 1]$ that starts at time $t = 0$ in the slow model (dotted line) and the fast model (solid line). The parameter values are $\sigma_u = 1$ (standard deviation of noise traders' order flow), $\sigma_v = 1$ (standard deviation of innovations in the asset value), $\sigma_e = 1$ (standard deviation of noise in news), and $\Sigma_0 = 1$ (variance of asset value conditional on information available at date 0).

5.4 Figure 3: cumulative price impact at different horizons

Read:

- At very short horizons, cumulative price impact is close to zero for the slow trader and strictly positive for the fast trader.
- The fast trader's trade therefore predicts immediate subsequent price changes.
- But the gap narrows at longer horizons because both fast and slow traders still exploit long-run valuation information.
- The paper explicitly notes that for the parameterization used in Figure 3, the fast trader still earns **86.75%** of his profit from the value-trading component.

5.5 Figure 4: infrequent news and dealer latency

Read:

- The figure introduces two extra objects:
 - m : the number of trading rounds per news interval,
 - ℓ : the dealer's latency.
- A larger ℓ means the trader has more chances to trade before the news is incorporated.
- This makes the notion of speed more realistic: it is not just a binary event but can reflect the length of the trader's head start.

5.6 Figure 5: news informativeness and speed

Read:

- The figure compares a stock with high news informativeness (**stock H**, $\nu = 2$) to one with low news informativeness (**stock L**, $\nu = 0.5$), holding the news frequency fixed at $m = 4$.
- When the trader is slow, illiquidity is lower for the high-news stock: roughly **1.10** for stock H versus **1.34** for stock L.
- When the trader is fast, illiquidity, value-trading intensity, news-trading intensity, and participation vary across rounds:
 - before news arrival, the dealer faces more adverse selection;
 - after news arrival, that adverse selection falls.
- The average predictions of the baseline model still survive:
 - average illiquidity is higher when the trader is fast than when he is slow,
 - average participation and news-trading intensity are higher for the stock with more informative news.
- The figure also quantifies the profit-composition result: for the illustrated calibration, news trading accounts for about **33.38%** of total profit in stock H, versus about **19.64%** in stock L.

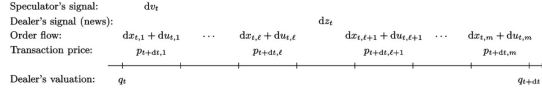


Figure 4. Timing of events during $[t, t + dt]$ for arbitrary news frequency, m , and dealer's latency, ℓ .

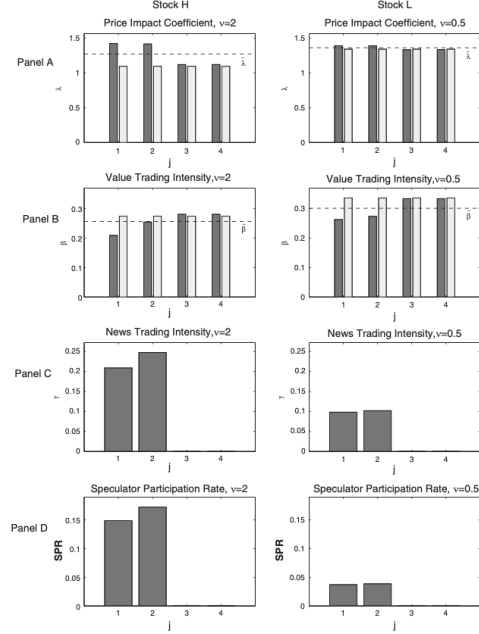


Figure 5. News informativeness and speed. We plot equilibrium values of various variables of interest in a given interval $[t, t + dt]$ in each trading round j for two different stocks, H (plots on the left) and L (plots on the right). The news frequency is $m = 4$ for each stock. News informativeness is higher for stock H than for stock L ($v = \frac{1}{\sigma_v} = 2$ versus $v = \frac{1}{\sigma_v} = 0.5$). For each stock, we show equilibrium values of $\lambda_{t,j}$ (illiquidity), $\beta_{t,j}$ (value trading intensity), $\gamma_{t,j}$ (news trading intensity), and $SPR_{t,j} = \frac{\text{Var}(dx_{t,j})}{\text{Var}(dx_{t,j}) + \text{Var}(du_{t,j})}$ in each trading round $j \in \{1, 2, 3, 4\}$ when the dealer's latency is $\ell = 2$ (left dark bars) or $\ell = 0$ (right light bars). The horizontal dotted lines correspond to the average value of the relevant variable over the four trading rounds when the speculator is fast (e.g., $\bar{\lambda} = (\sum_{j=1}^4 \lambda_{t,j})/4$) in the fast model. The other parameter values are $\sigma_u = 1$ (standard deviation of noise traders' order flow), $\sigma_v = 1$ (standard deviation of innovations in the asset value), and $\Sigma_0 = 1$ (variance of asset value conditional on information available at date 0).

6 Short summary

- **Method.** The paper develops a continuous-time Kyle-style model in which the dealer receives public news and the informed trader's private signals forecast both long-run value and short-run news.
- **Key mechanism.** If the trader is fast enough to trade before the dealer updates quotes to current news, his optimal trade acquires a separate news-trading component.
- **Main implications.** Fast traders contribute more to volume, have less autocorrelated trades, and predict short-horizon returns more strongly. Yet they still earn an important share of profits from longer-horizon value trading.
- **Market-quality result.** Speed worsens liquidity but leaves the overall speed of price discovery unchanged.
- **Cross-sectional result.** More informative news raises a fast trader's activity and the share of profits coming from short-run bets, while making speed more valuable.

7 Conclusion

7.1 Core conclusion

- The paper shows that speed creates a qualitatively new equilibrium behavior: **news trading**.

- It also shows that observing very short-horizon predictive trading is not enough to conclude that profits are earned only from very short horizons.
- Finally, it shows that speed and public-information precision jointly shape both HFT behavior and market liquidity.

7.2 Economic intuition

- A fast trader can buy before good news moves the price and sell before bad news moves the price.
- But doing so reveals information faster, so he tones down his slower-moving value bet.
- The dealer then learns more from trades and less from subsequent news.
- That is why order flow becomes more important, news becomes less important, and liquidity worsens even though overall pricing efficiency does not change.

7.3 Takeaway

This paper provides a theoretical foundation for interpreting directional HFT. The key message is subtle: high-frequency traders may look extremely short-term in their order flow, yet still be doing economically meaningful long-horizon information trading. Speed changes not only how much they trade, but also the composition of their bets, the way prices react to trades versus news, and the role of news informativeness in shaping liquidity and profitability.